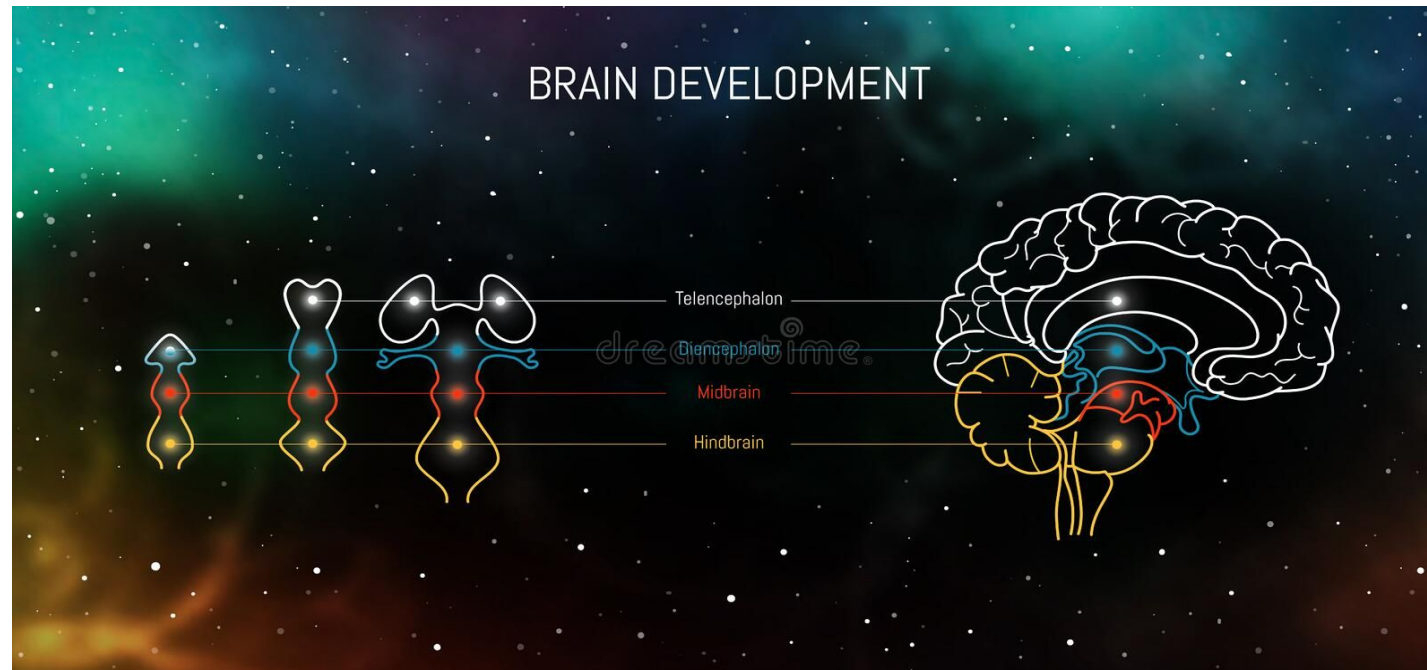
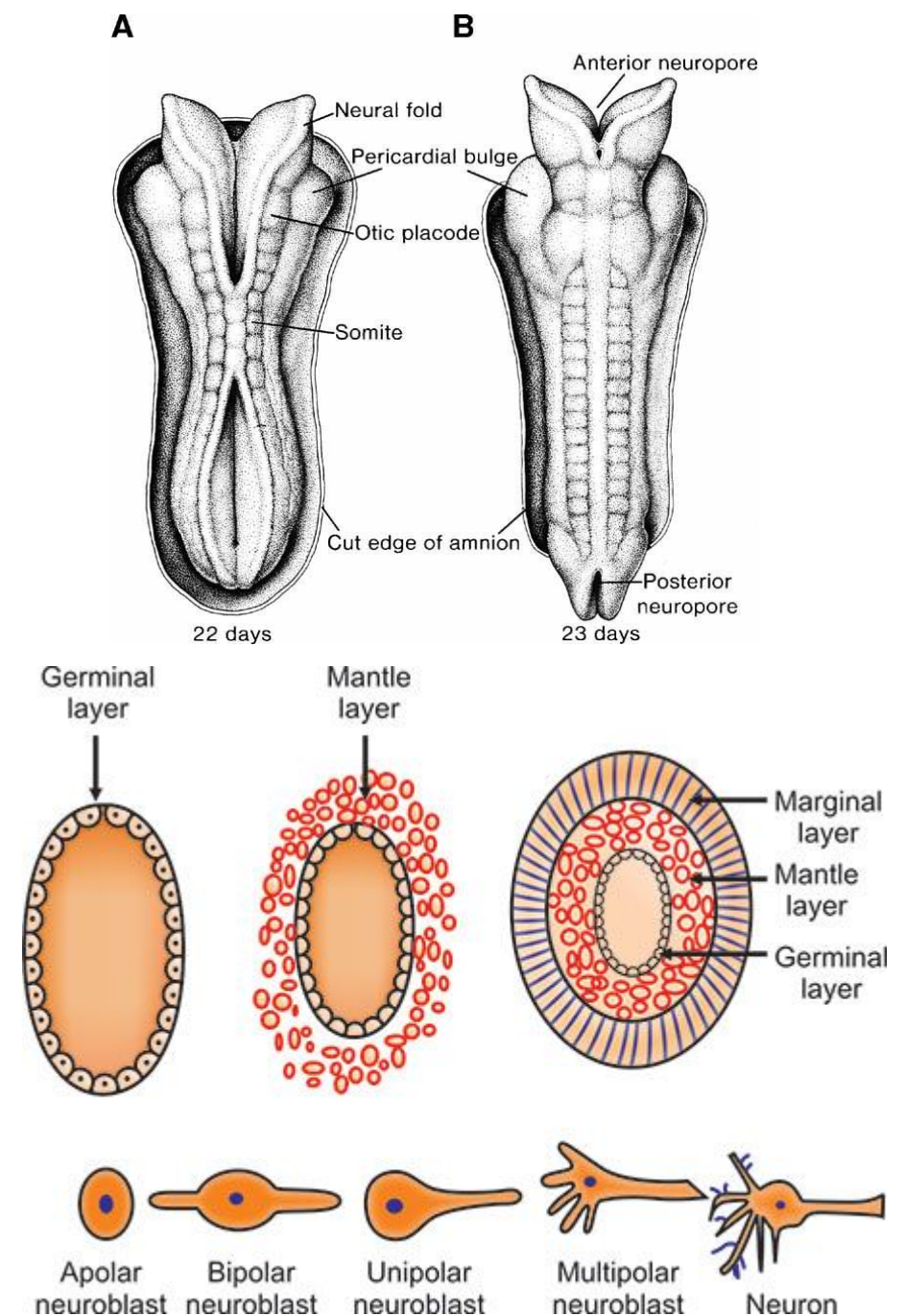


Overview of the Development of Nervous System

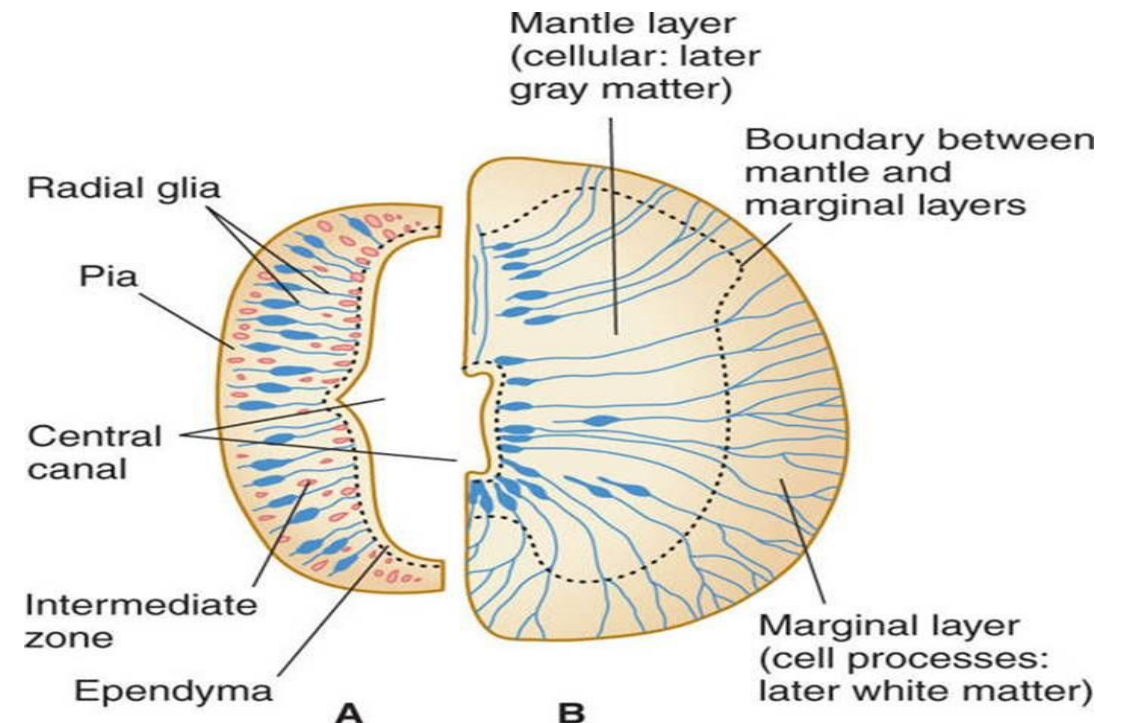
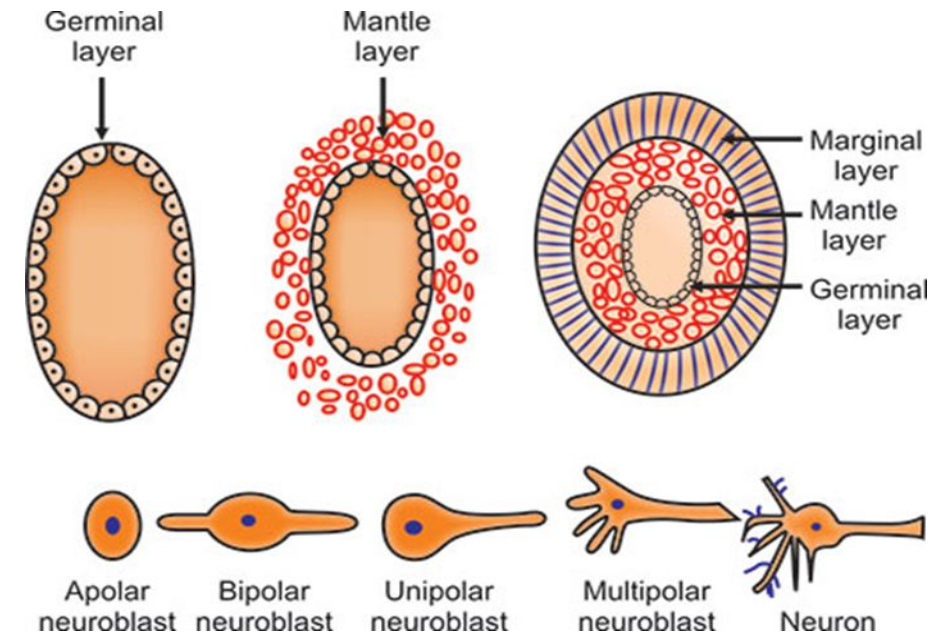


Introduction:

- The wall of a freshly-closed neural tube consists of neuroepithelial cells.
- After the closure of neural tube, neuroepithelial cells give rise to another cell type called **neuroblasts**.
- The neuroblasts are characterized by a pale, large, spherical nucleus with a prominent nucleolus.
- The neuroblasts form a **mantle layer** around neuroepithelial layer.

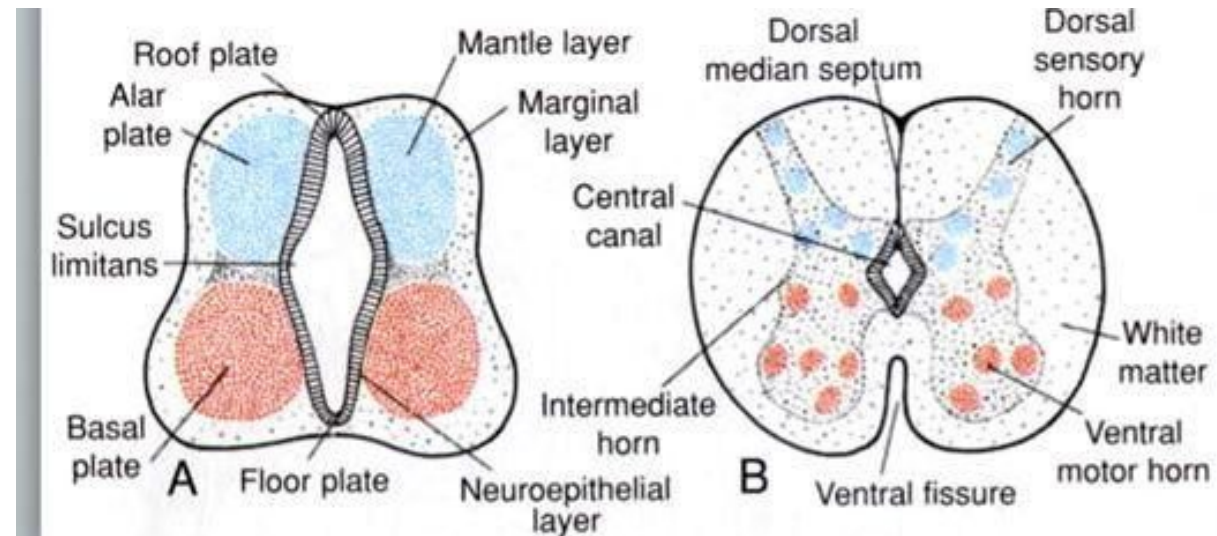
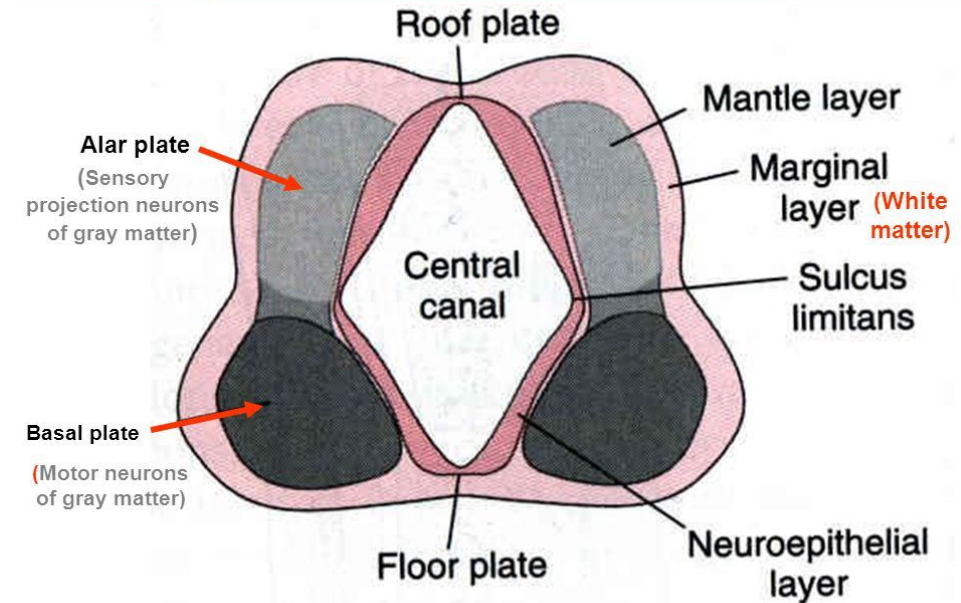


- The **mantle layer** later forms **gray matter** of spinal cord.
- The **marginal layer** is the **outermost layer** of the spinal cord.
- It consists of nerve fibers coming from neuroblasts in the mantle layer.
- The **marginal layer** forms **white matter** of the spinal cord.

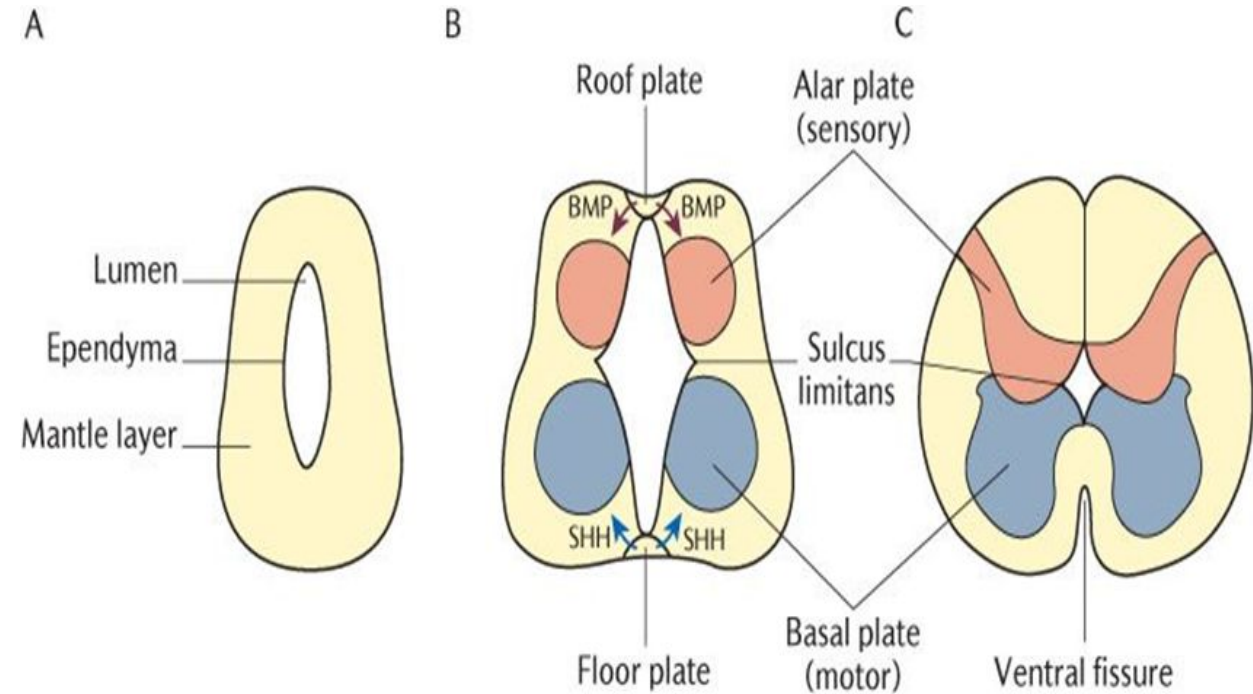


- The continuous addition of neuroblasts to the mantle layer, produces a ventral and a dorsal thickening on each side of neural tube.
- The **ventral thickenings** are called **basal plates** which **form motor areas** of spinal cord.
- The **dorsal thickenings** are called **alar plates**, **form sensory areas**.
- The basal and alar plates of each side are separated by a longitudinal groove called sulcus limitans.

Formation of basal and alar plates of neural tube

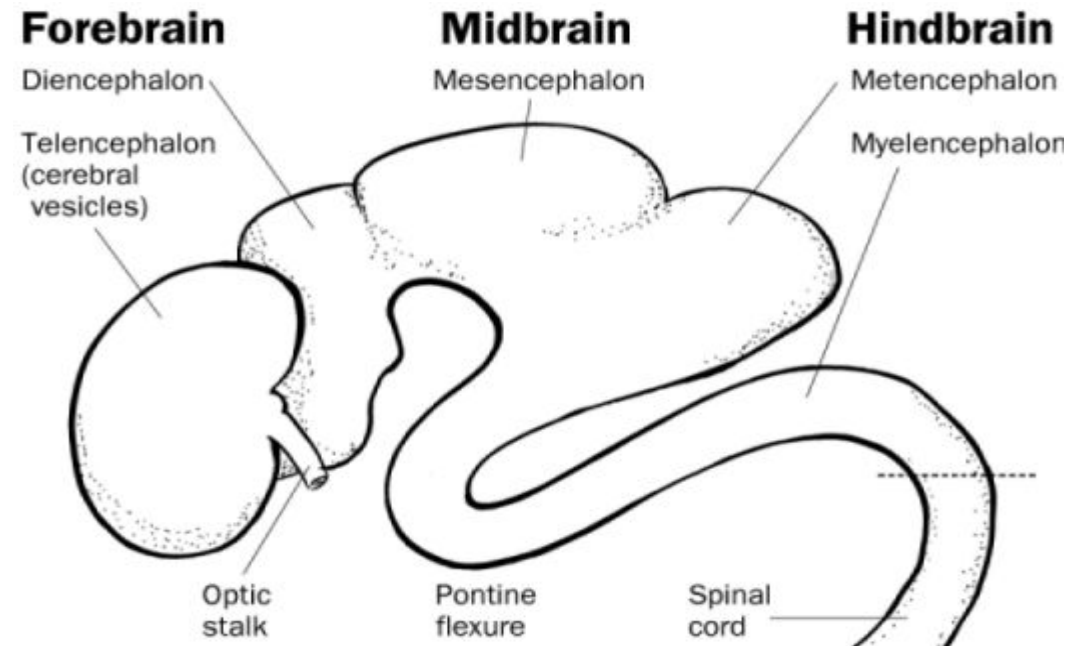


- The **dorsal and ventral** midline portions of neural tube are called **roof and floor plates** respectively.
- They do not contain neuroblasts.
- The roof & floor plates mainly serve as pathways for nerve fibers crossing from one side to the other.

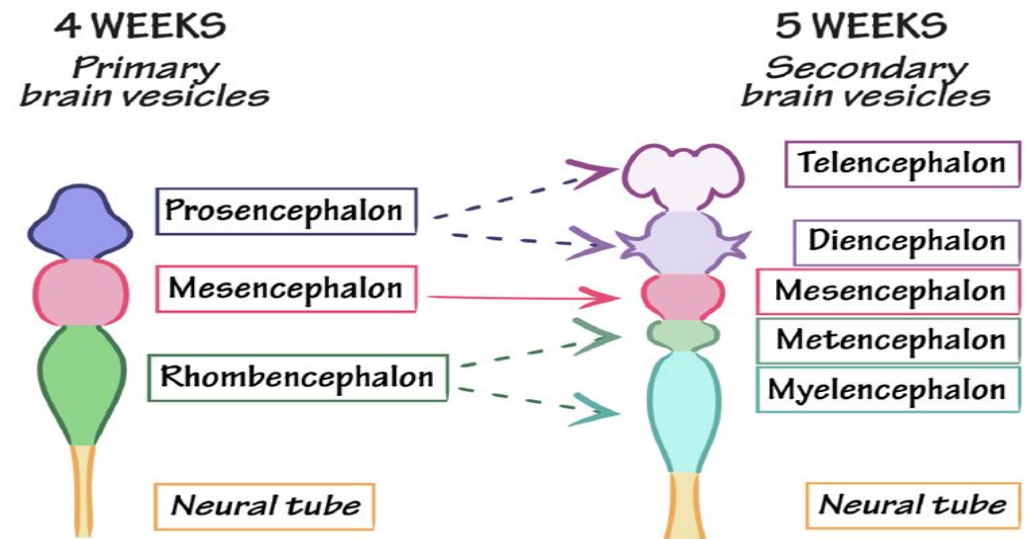


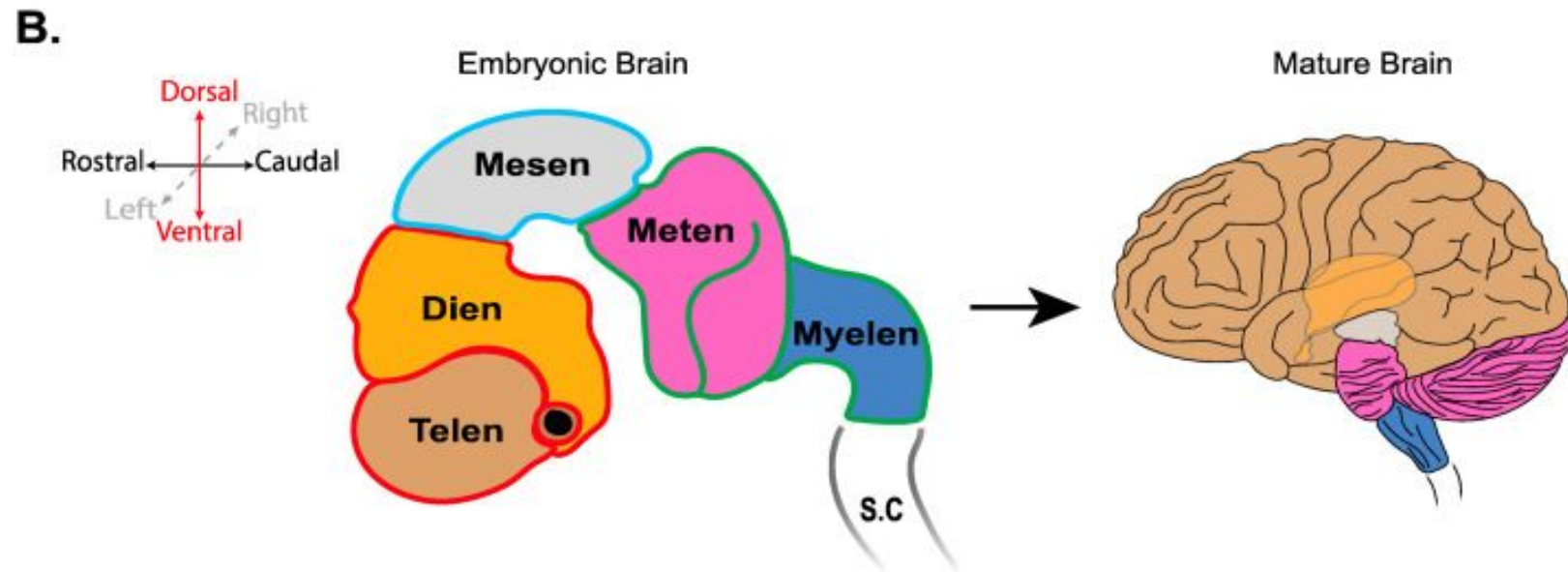
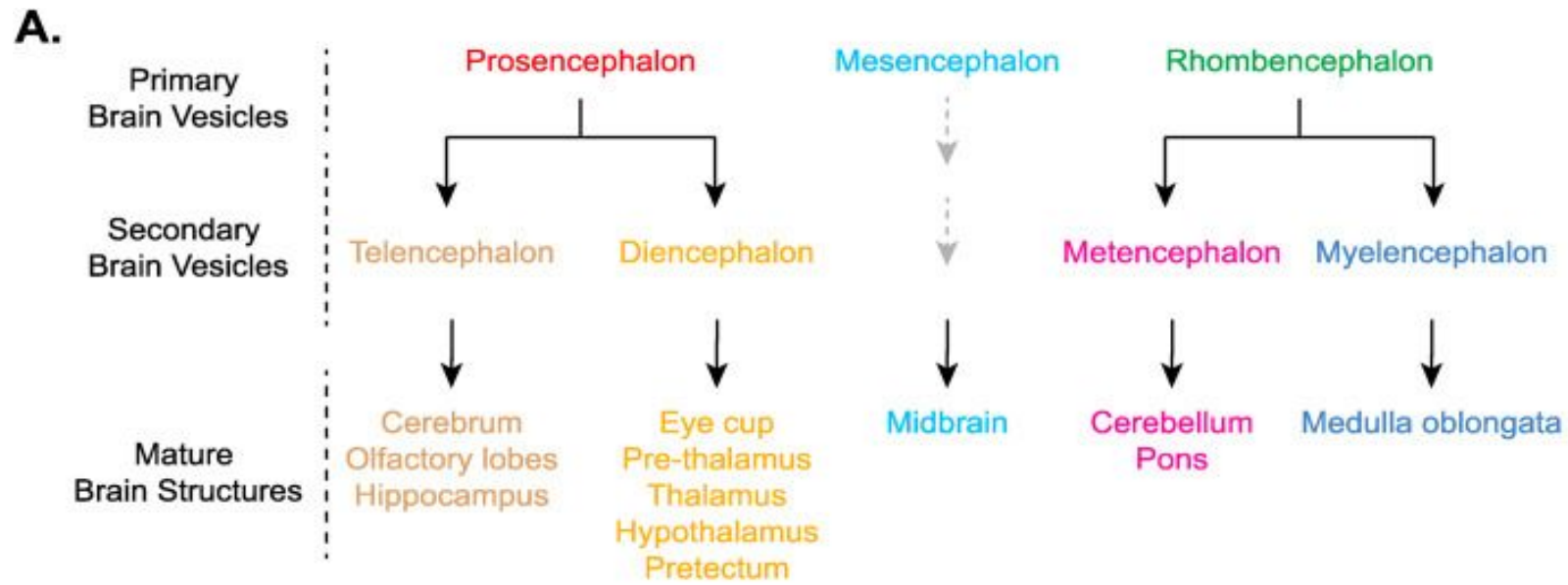
Development of Brain:

- Once the neural tube has closed, **3 primary vesicles**, the **forebrain**, **midbrain**, and **hindbrain vesicles**, complete their development.
- The **forebrain vesicle** will form **forebrain (prosencephalon)**.
- The **midbrain vesicle** will form **midbrain (mesencephalon)**.
- The **hindbrain vesicle** will form **hindbrain (rhombencephalon)**.
- By fifth week, the forebrain and hindbrain vesicles divide into two secondary vesicles.
- The **forebrain vesicle forms (1) telencephalon**, with its primitive cerebral hemispheres, & **(2) diencephalon** which develops optic vesicles.

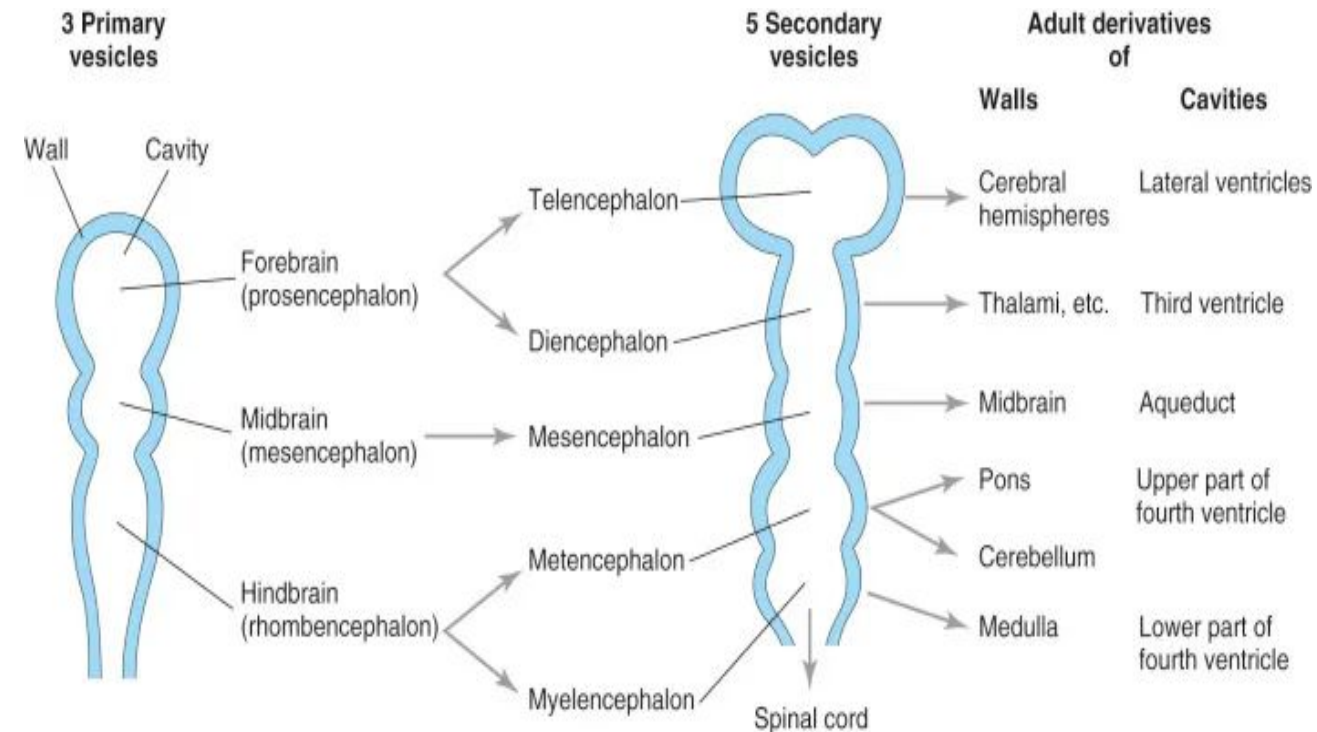
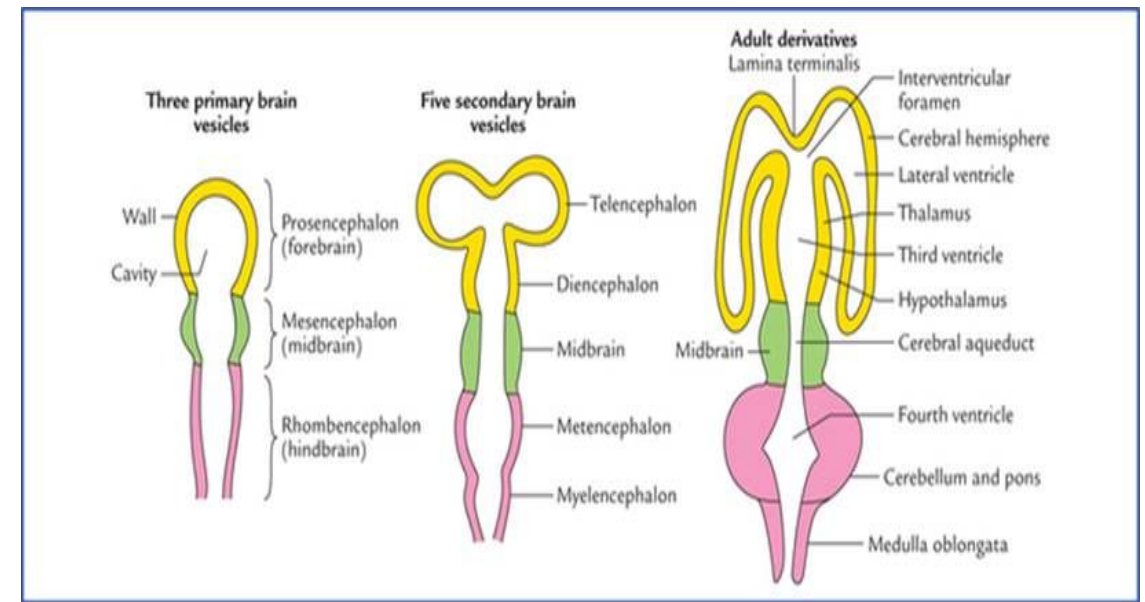


CORONAL VIEW

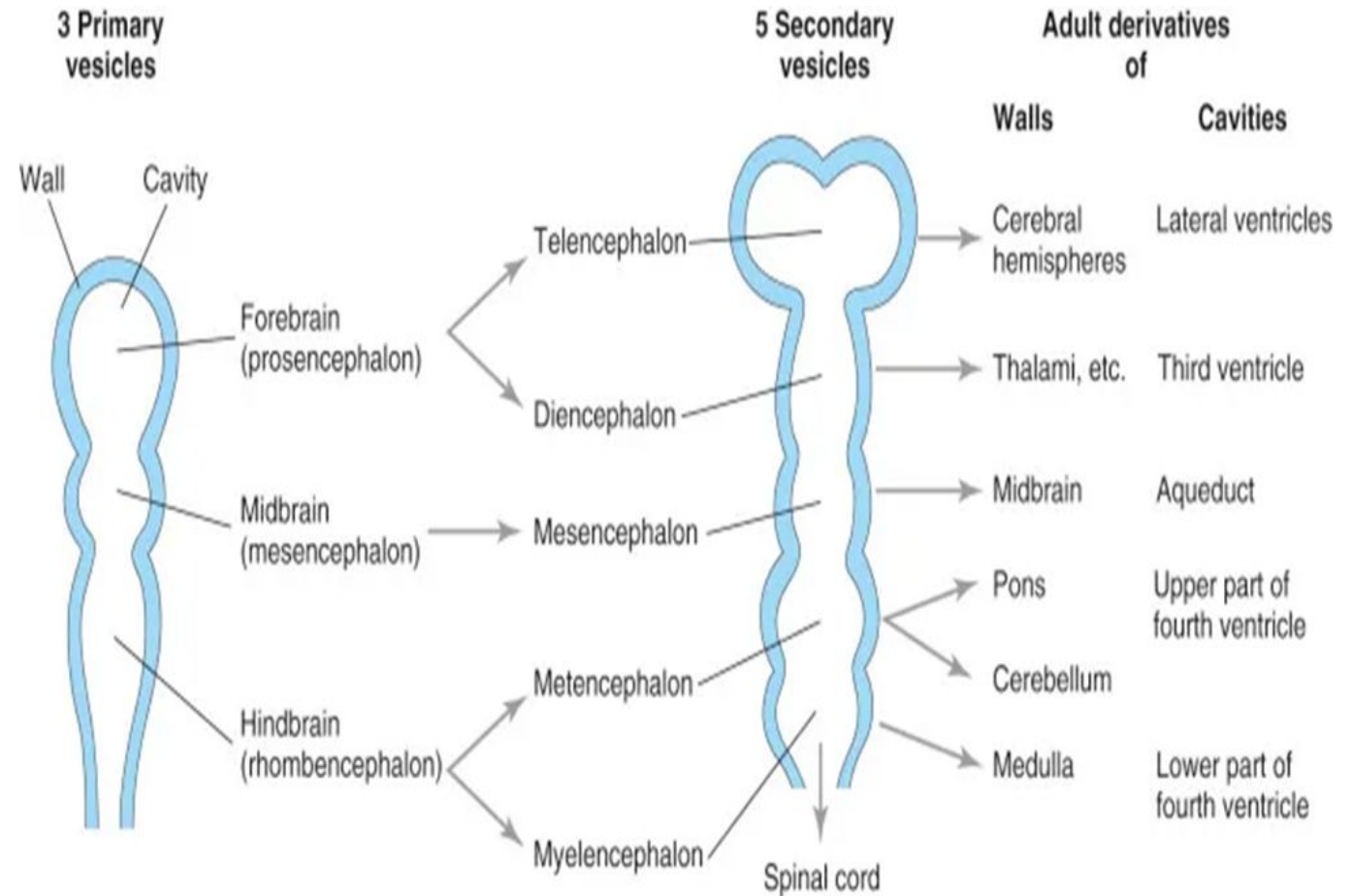




- The **hindbrain vesicle** forms **(1) metencephalon**, the future pons and cerebellum, & **(2) myelencephalon** or medulla oblongata.
- The basic pattern in ventricular system is now established.
- The cavity in each cerebral hemisphere is known as lateral ventricle.
- The cavity of diencephalon is called third ventricle.

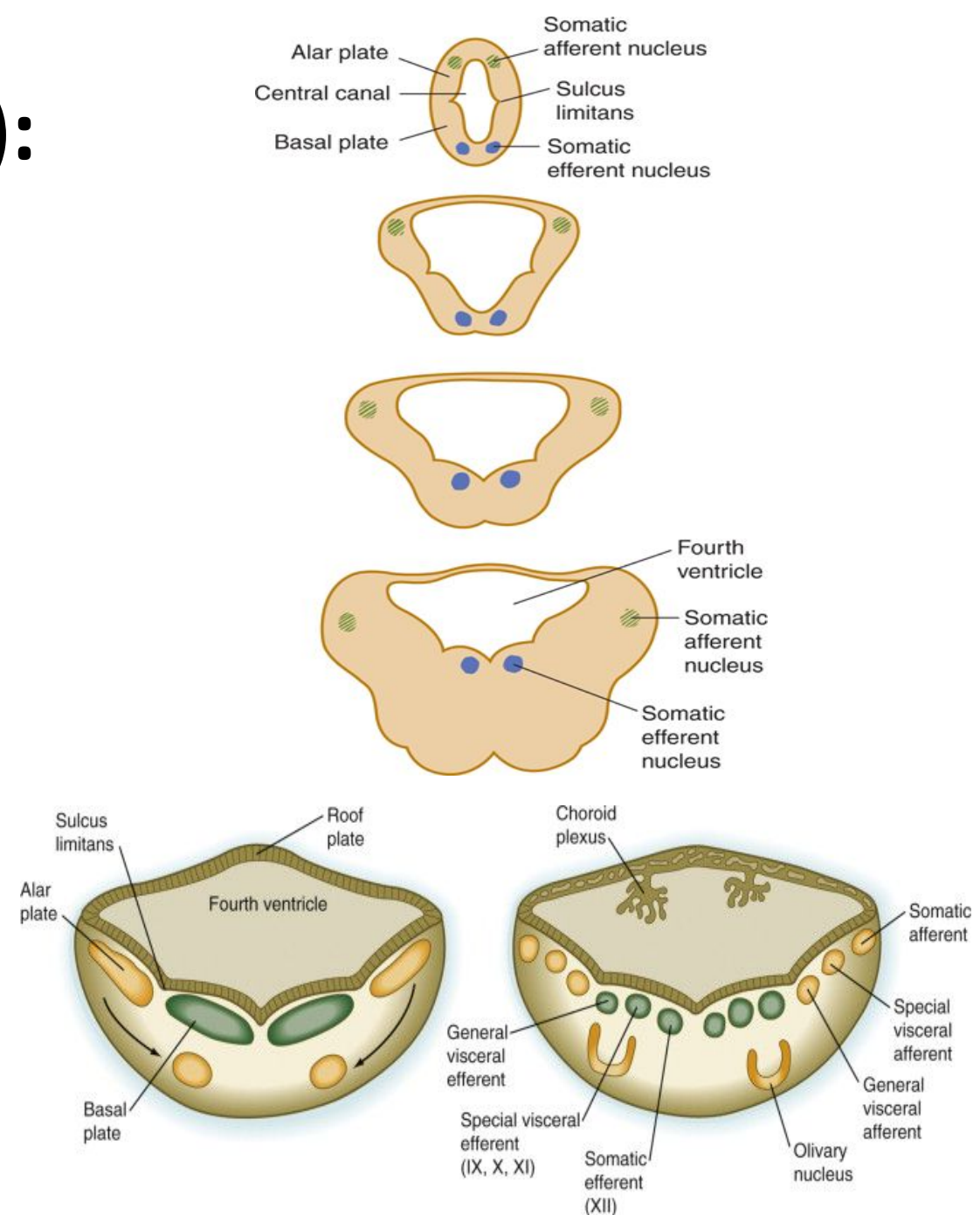


- With continuous growth, the **cavity of midbrain vesicle** becomes small and forms **cerebral aqueduct/ aqueduct of Sylvius**.
- The **cavity of hindbrain vesicle** forms **fourth ventricle**, which is continuous with the central canal of spinal cord.
- The lateral ventricles communicate with the third ventricle through interventricular foramina.
- The ventricular system and central canal of spinal cord is lined with ependyma and filled with cerebrospinal fluid.

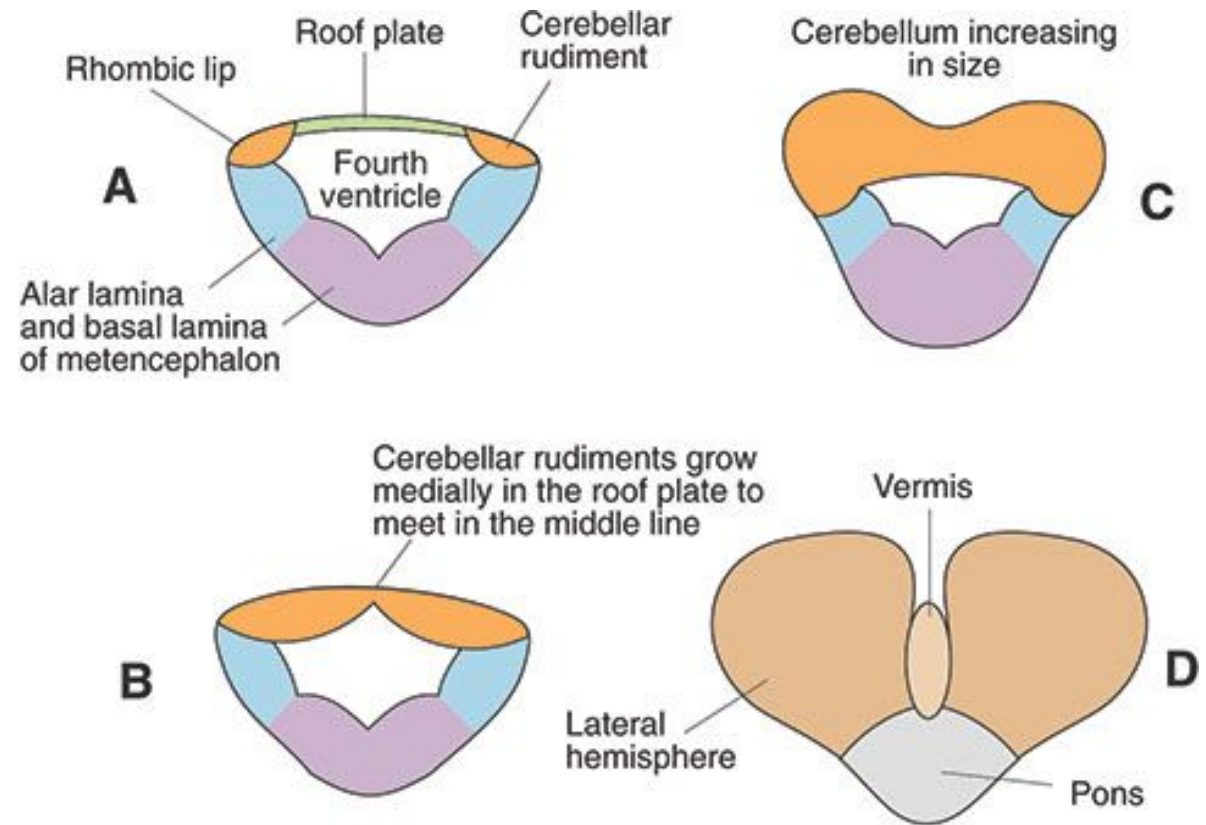


Rhombencephalon (hindbrain):

- The walls of hindbrain vesicle initially shows typical organization in the neural tube.
- The anterior thickenings are known as basal plates, and
- The posterior thickenings are called alar plates.
- These plates are separated by sulcus limitans.
- The basal plates contain motor neurons and alar plates contain sensory neurons.
- As development proceeds, the lateral walls are moved laterally with the expanding ventricles.
- Thus, alar plates come to lie lateral to the basal plates.

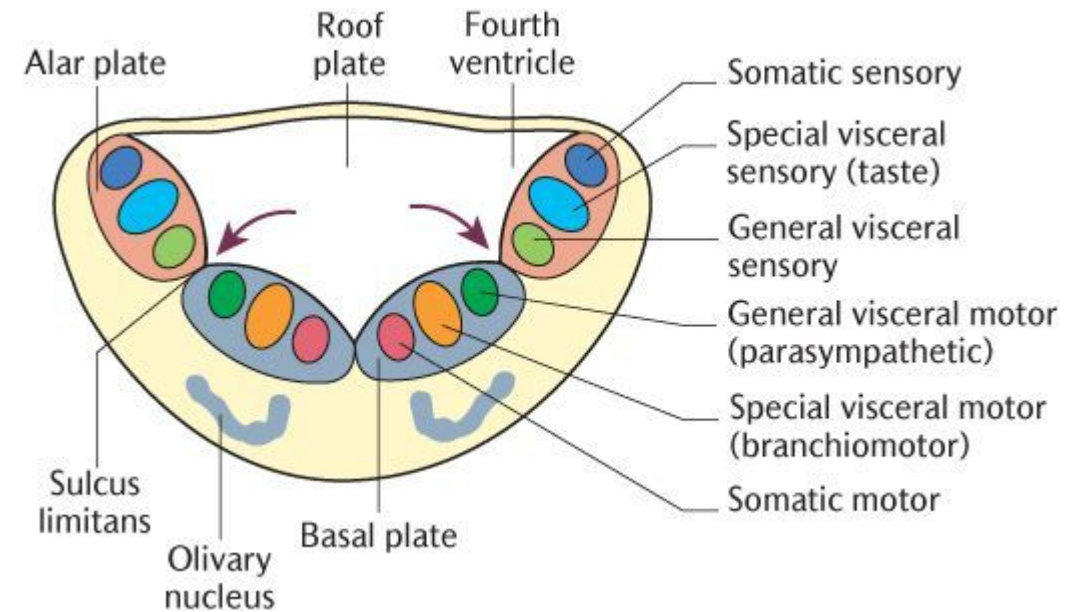
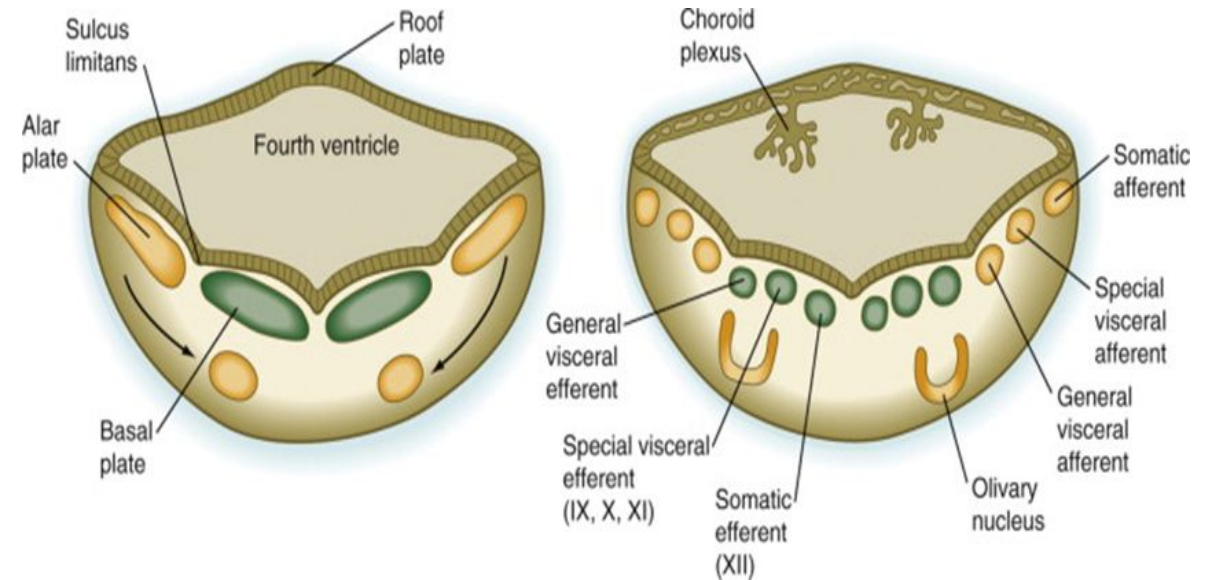


- The hindbrain vesicle is characterized by the formation of cerebellum, a coordination center for posture and movement, &
- Pons, the pathway for nerve fibers between spinal cord and cerebral & cerebellar cortices.

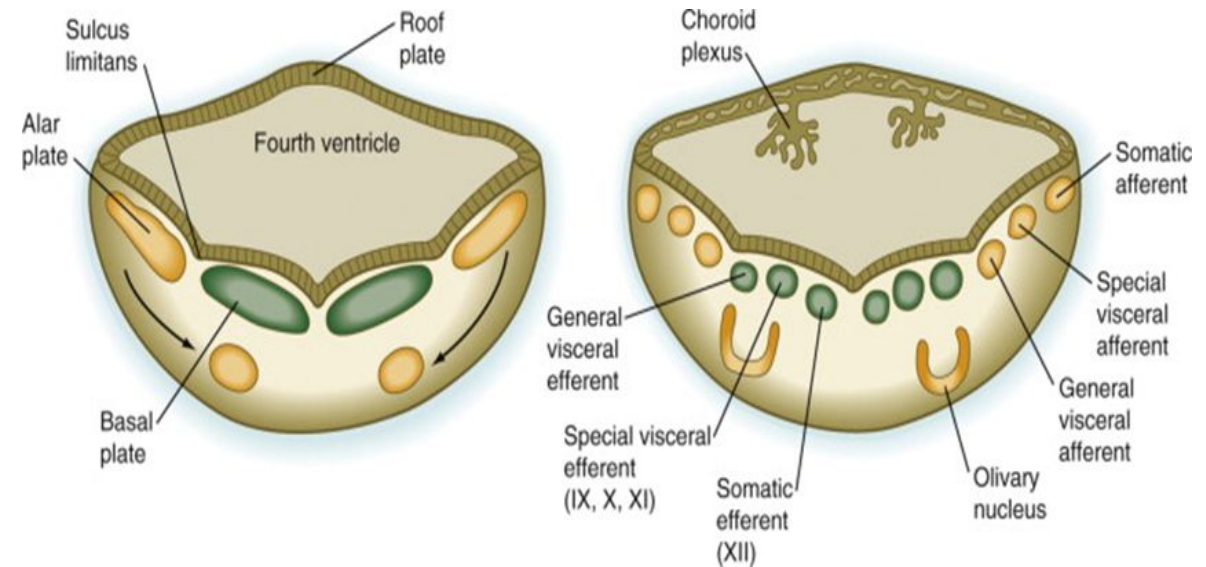


Myelencephalon:

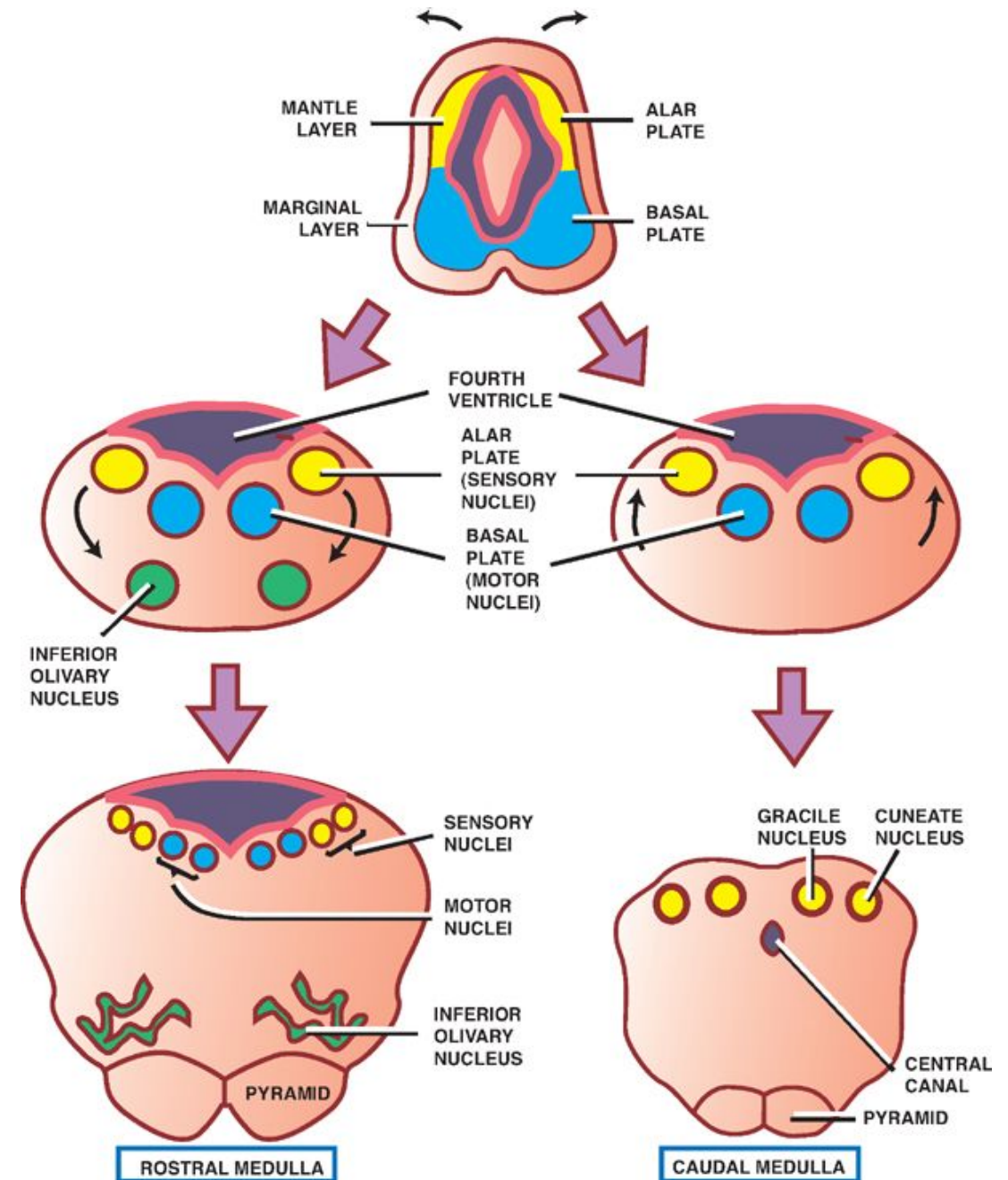
- The neurons of basal plates form motor nuclei of cranial nerves IX, X, XI & XII and are situated in the floor of fourth ventricle medial to sulcus limitans.
- the neurons of the alar plate form sensory nuclei of cranial nerves V, VIII, IX & X, and gracile & cuneate nuclei.
- Some cells of alar plate migrate ventrolaterally to form olivary nuclei.



- The roof plate becomes stretched into a thin layer of ependymal tissue.
- The vascular mesenchyme lying in contact with the outer surface of roof-plate forms pia mater, & two layers together form tela choroidea.
- Vascular tufts of tela choroidea project into the cavity of fourth ventricle to form choroid plexus.
- Between fourth & fifth months, local resorption of roof plate occurs forming paired lateral foramina, foramina of Luschka, and a median foramen of Magendie.

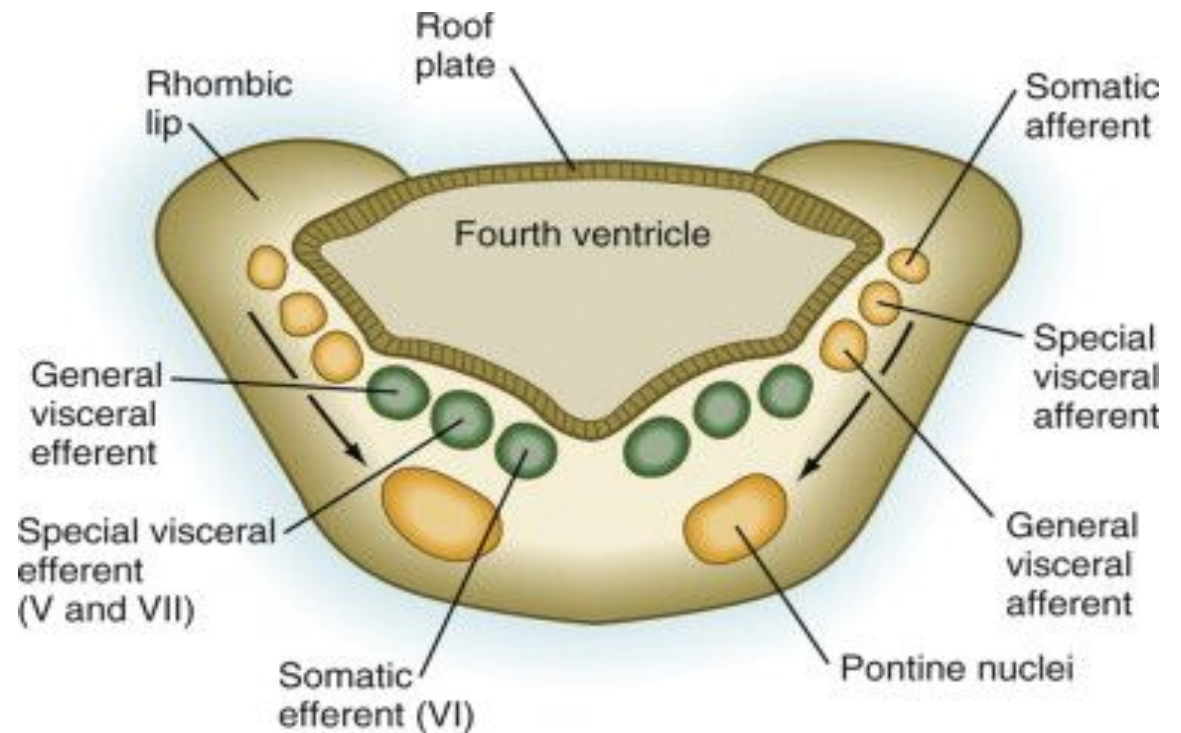


- The floor plate remains narrow and forms the region of median sulcus.
- In marginal layer on the anterior aspect of medulla, descending axons from neurons in the motor areas of cerebral cortex (precentral gyrus) produce prominent swellings called the pyramids.

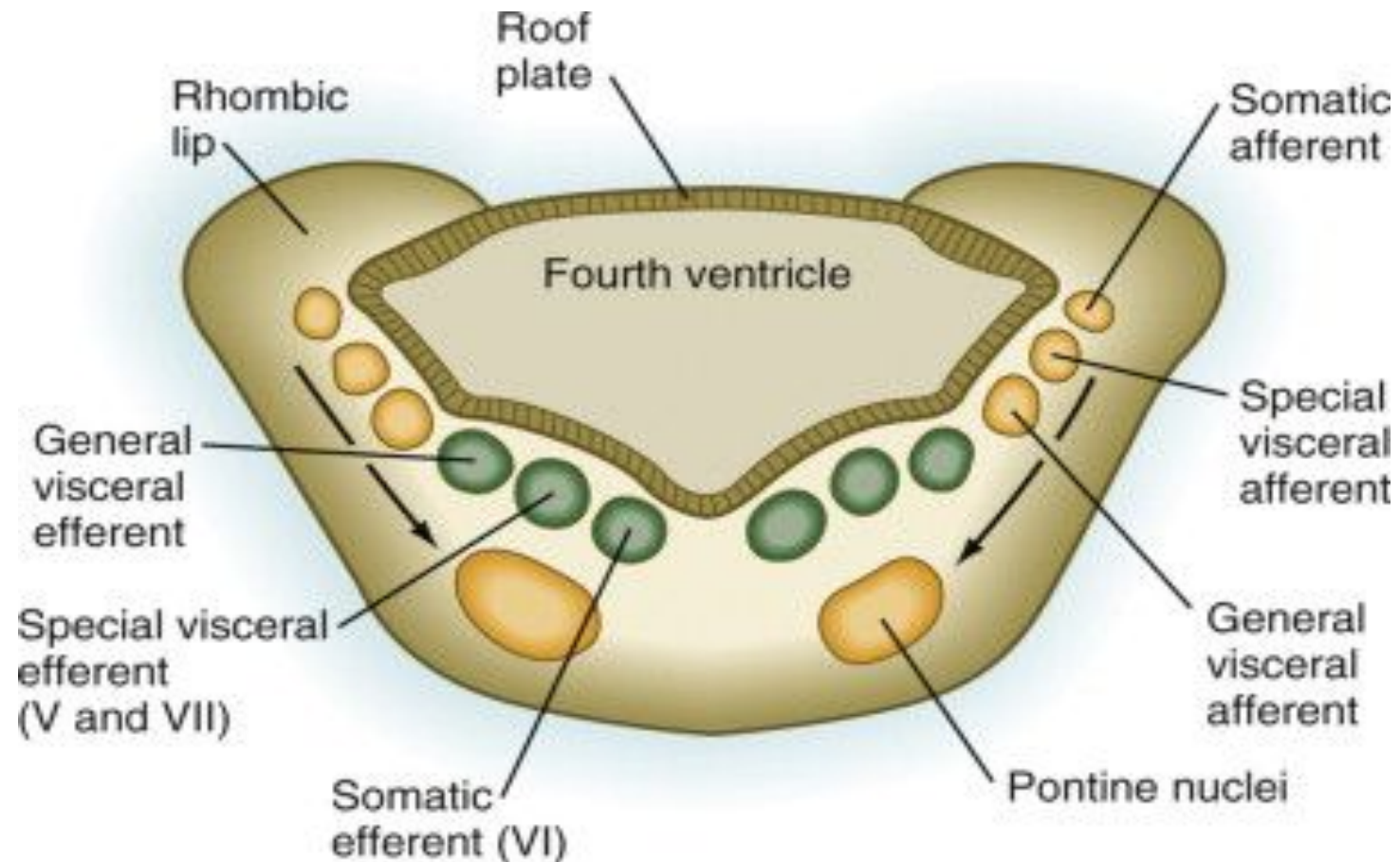


Metencephalon (Pons):

- The pons arises from anterior part of metencephalon, but it also receives a cellular contribution from alar part of the myelencephalon.
- The neurons of basal plates form motor nuclei of cranial nerves V, VI, & VII.
- The neurons of ventromedial part of each alar plate form main sensory nucleus of cranial nerve V, a sensory nucleus of nerve VII, and vestibular and cochlear nuclei of cranial nerve VIII.
- They also form pontine nuclei.

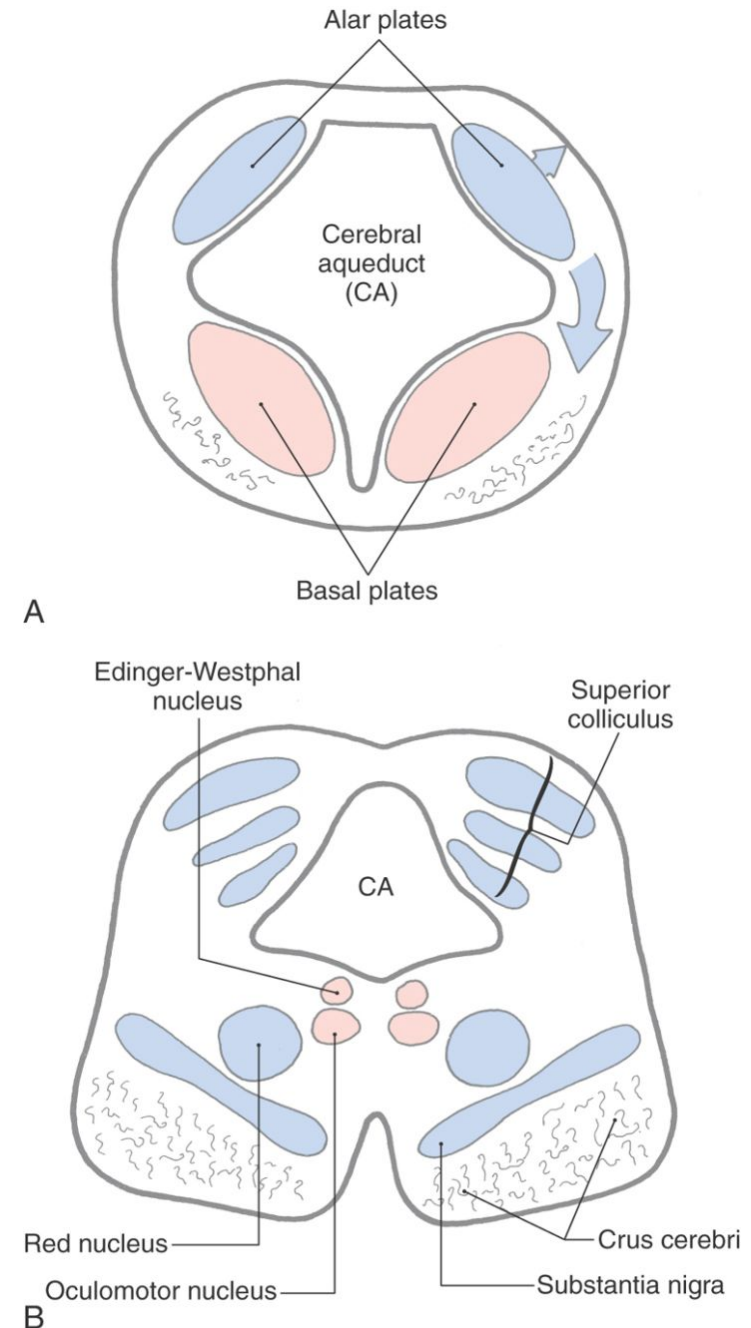


- The axons of pontine nuclei grow transversely to enter developing cerebellum of opposite side, thus forming transverse pontine fibers and middle cerebellar peduncles.



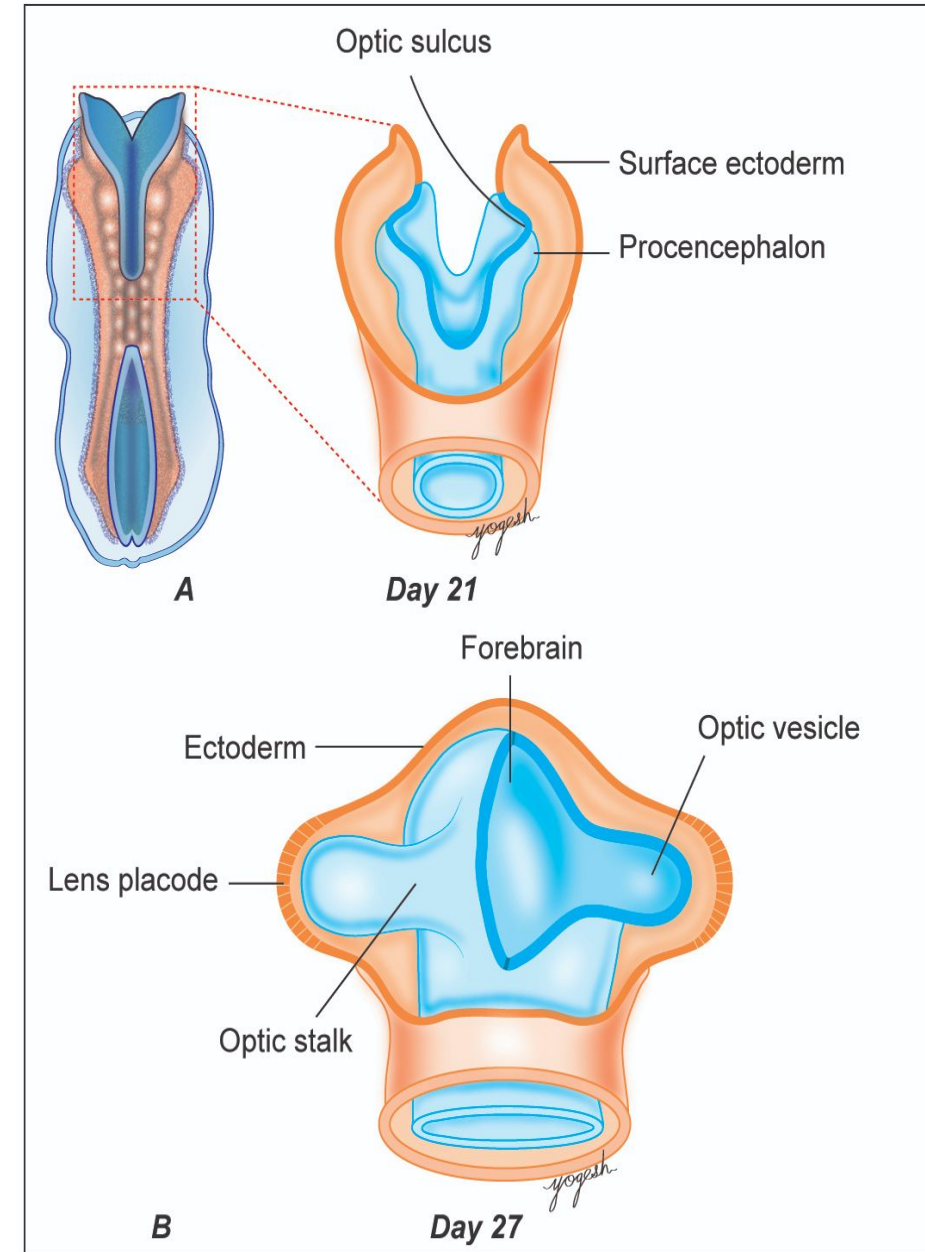
Mesencephalon:

- The midbrain (mesencephalon) develops from midbrain vesicle, the cavity of which is reduced to form cerebral aqueduct.
- The midbrain vesicle does not divide & resembles spinal cord with its basal efferent and alar afferent plates.
- The sulcus limitans separates alar plate from basal plate on each side.
- The neuroblasts in basal plates will differentiate into neurons forming nuclei of third and fourth cranial nerves and possibly Edinger-Westphal nucleus, red nuclei, substantia nigra, and reticular formation.
- Here, the alar plates form anterior and posterior colliculi as visual and auditory reflex centers respectively.
- The two alar plates and the original roof plate form the tectum.

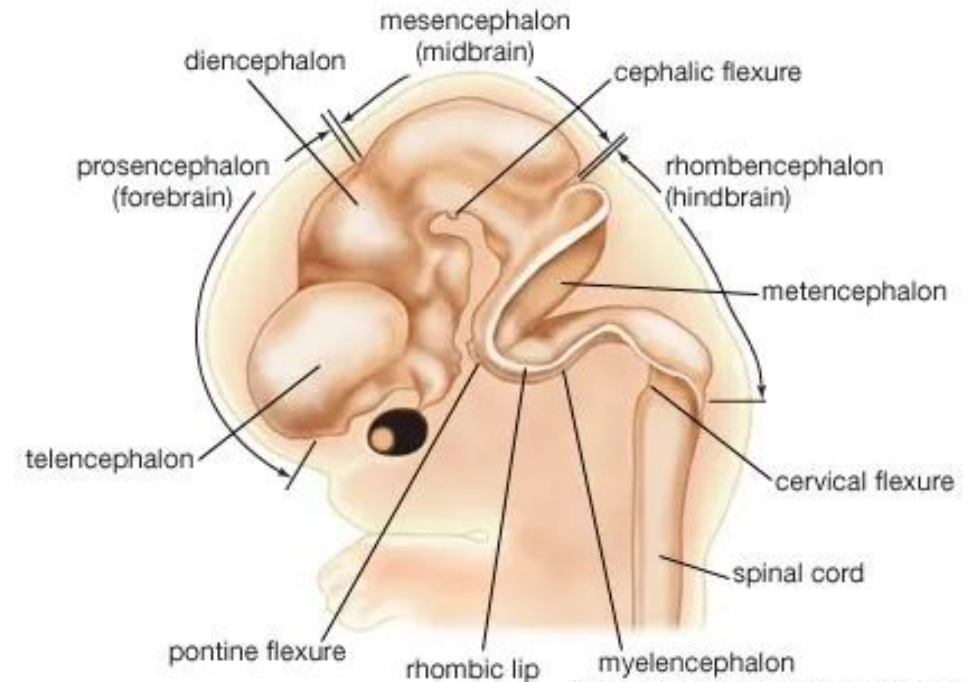


Prosencephalon (forebrain):

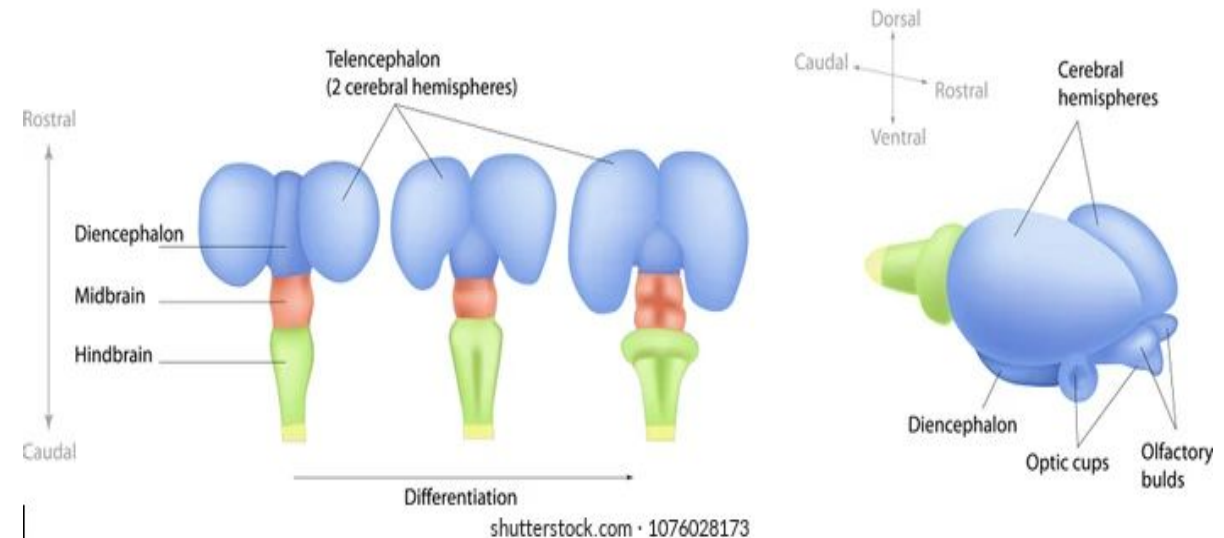
- The forebrain develops from forebrain vesicle.
- At an early stage, a lateral diverticulum appears on each side of forebrain, the optic vesicle.
- The part of forebrain that is rostral to optic vesicle is called **telencephalon** and the remainder is known as **diencecephalon**.
- The optic vesicle and its stalk finally form the retina and optic nerve respectively.
- The prosencephalon also subdivides into diencephalon posteriorly and telencephalon anteriorly.



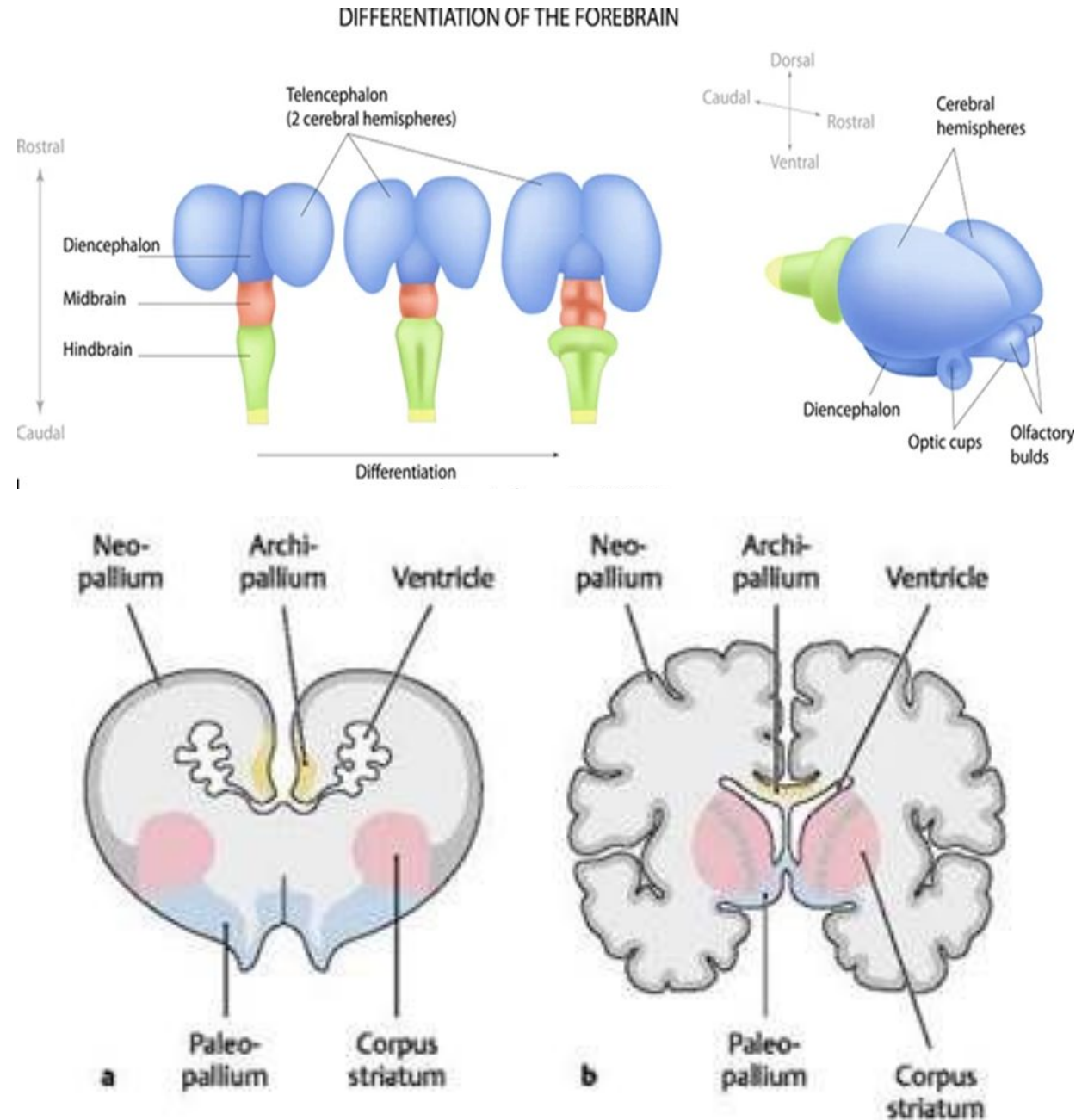
- The telencephalon forms cerebral hemispheres, and the diencephalon forms the optic cup and stalk, pituitary, thalamus, hypothalamus, & pineal body (epiphysis).
- The diencephalon consists of a thin roof-plate and a thick alar plate in which thalamus and hypothalamus develop.
- It participates in the formation of pituitary gland, which also develops from Rathke's pouch.



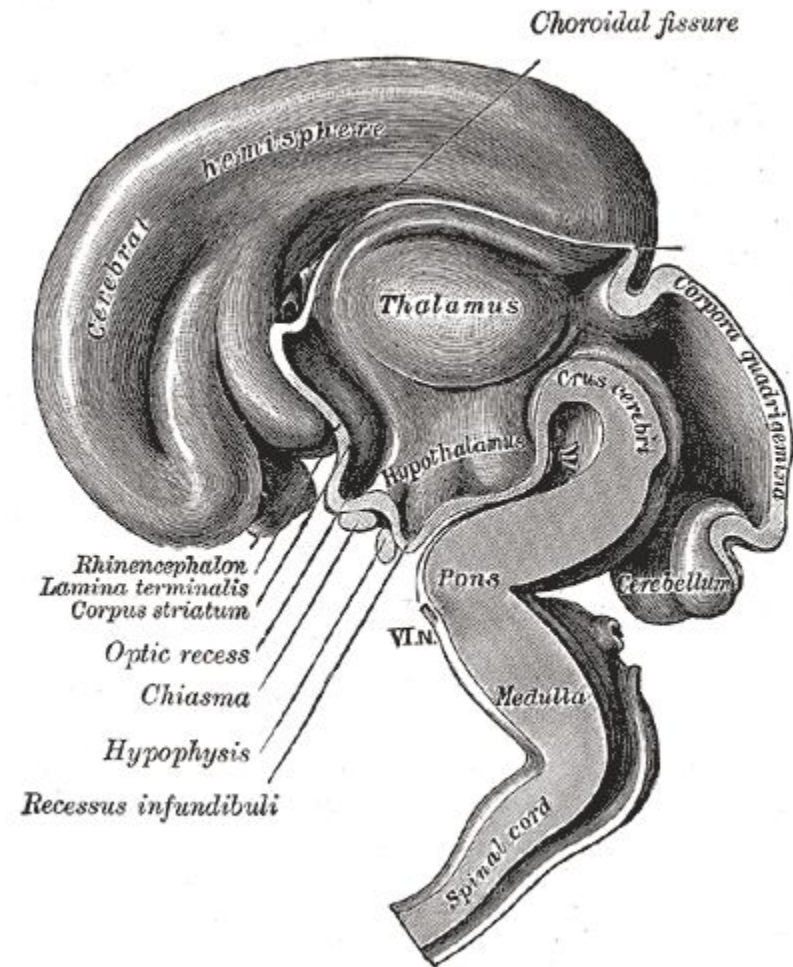
DIFFERENTIATION OF THE FOREBRAIN

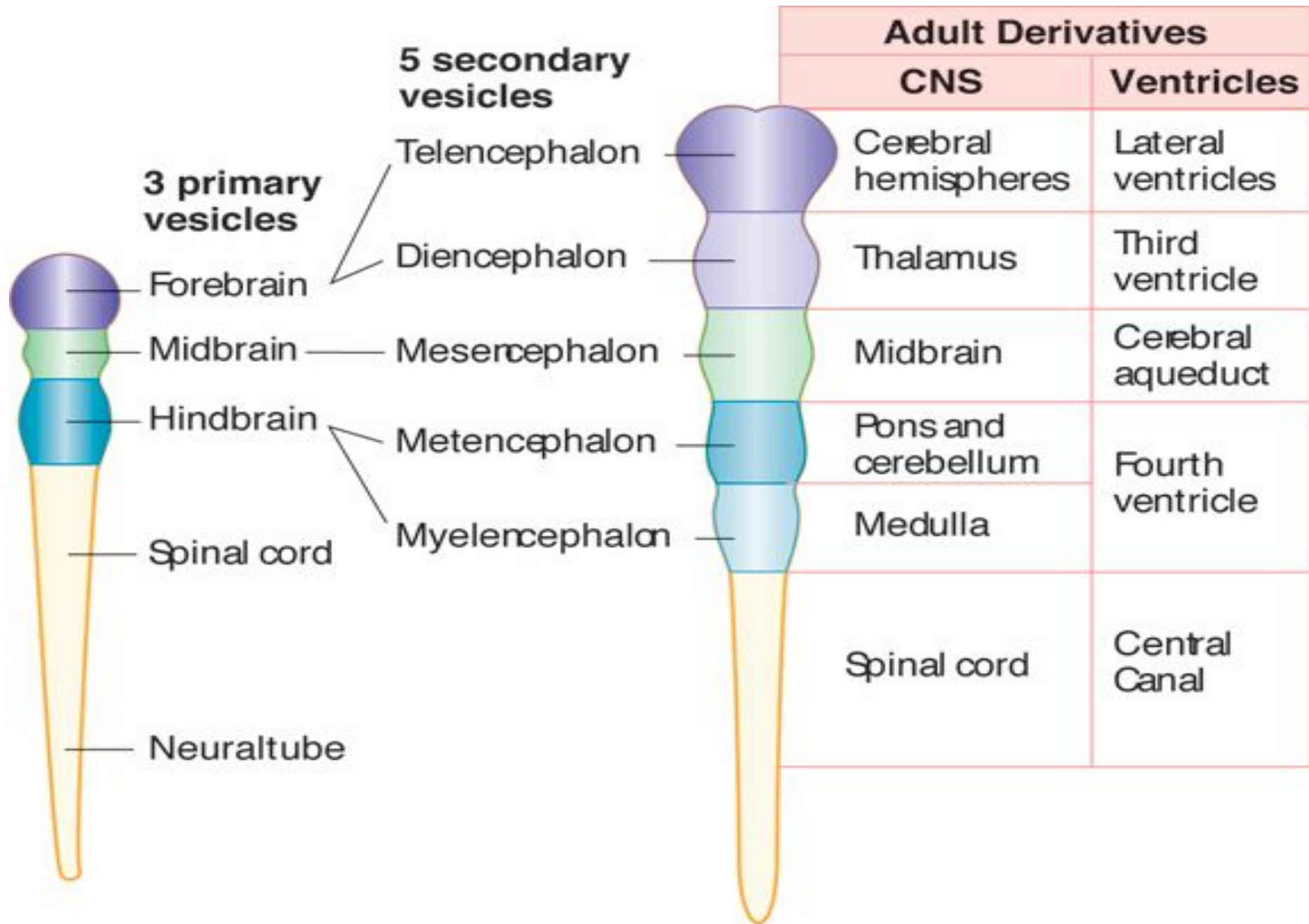


- The telencephalon consists of two lateral out-pocketings/evaginations, the cerebral hemispheres, and a median portion, lamina terminalis.
- The cerebral cortex develops from pallium.
- The pallium has three regions: paleocortex/paleopallium, archicortex/archipallium, & neocortex/neopallium.
- These small out-pocketings of cerebral hemispheres expand and cover the lateral aspect of diencephalon, mesencephalon, & metencephalon.
- Finally nuclear regions of telencephalon come in close contact with those of diencephalon.



- The lamina terminalis is used by commissures as a connection pathway for fiber bundles between right & left cerebral hemispheres.
- The largest commissure, corpus callosum, initially forms a small bundle in lamina terminalis.
- With continuous expansion of neopallium, this small bundle (of developing corpus callosum) extends anteriorly and then posteriorly, arching over the roof of diencephalon.

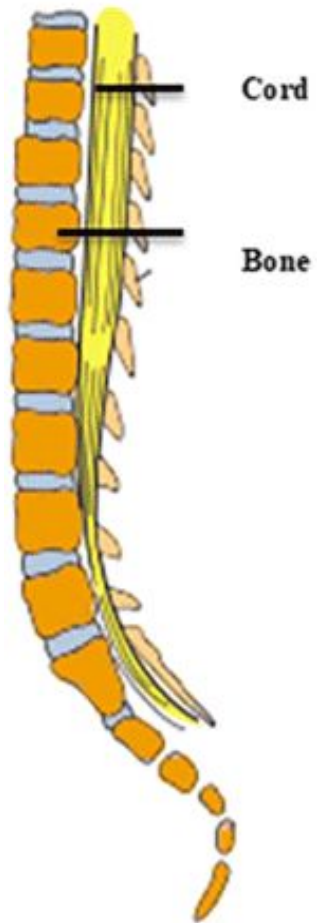




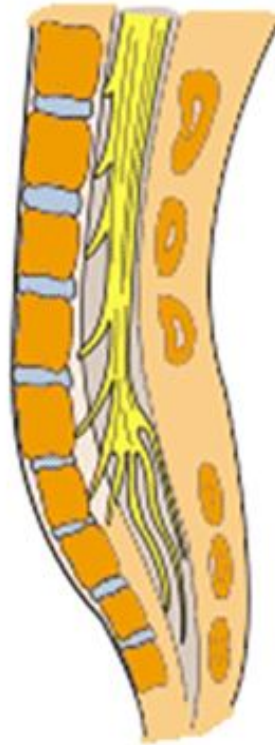
Neural Tube Defects (NTDs):

- **Anencephaly:** this condition occurs when neural tube fails to close in the cranial region and most of the brain fails to form.
- **Spina bifida:** this occurs when the closure of neural tube fails to occur anywhere from cervical to caudal region. It results in the imperfect fusion or non-union of the vertebral arches (*cleft vertebra*).
- **Spina bifida occulta:** in this, the bony defect is covered by skin and there are no neurological deficits.
- **Spina bifida cystica:** it is a severe abnormality in which neural tube fails to close, vertebral arches fail to form and neural tissue is exposed.
- Spina bifida may include meningocele and myelomeningocele. These are severe NTDs in which neural tissue or meninges protrude through the defect.
- **Meningocele:** in this case, only fluid-filled meninges protrude through the defect.
- **Myelomeningocele:** in this case, the neural tissue is included in the meningeal sac.

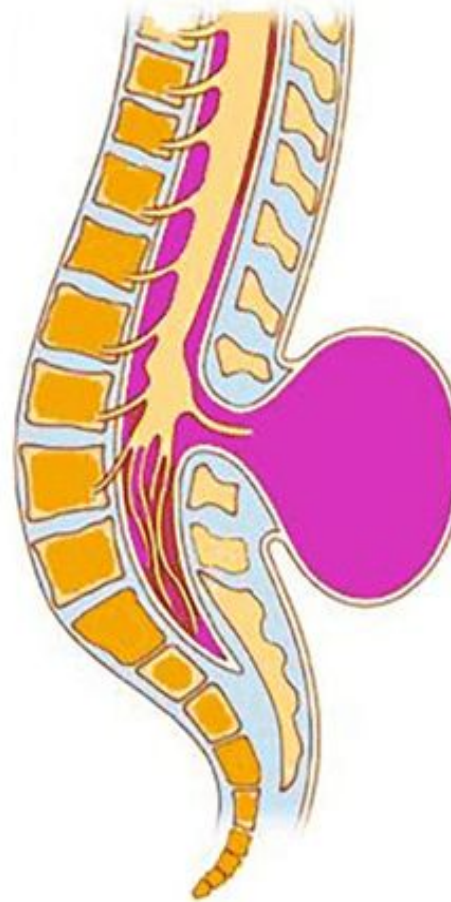
NTDS:



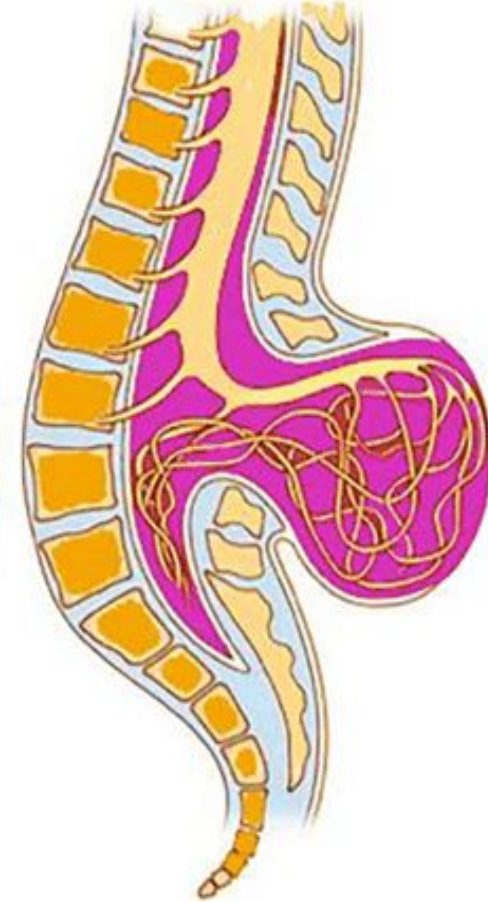
Normal Spine



Spina bifida
occulta



Meningocele



Myelomeningocele

References:

- **Clinical Neuroanatomy by Richard S. Snell,**
 - **8th edition**
 - **Page #: 502-512**